Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	**Aim: To understand the process of convolution over the image and apply over the classification problem**	
Experiment No: 4	**Date: 15/2/2026**	**Enrolment No: 92301733059**

Aim: To understand the process of convolution over the image and apply over the classification problem

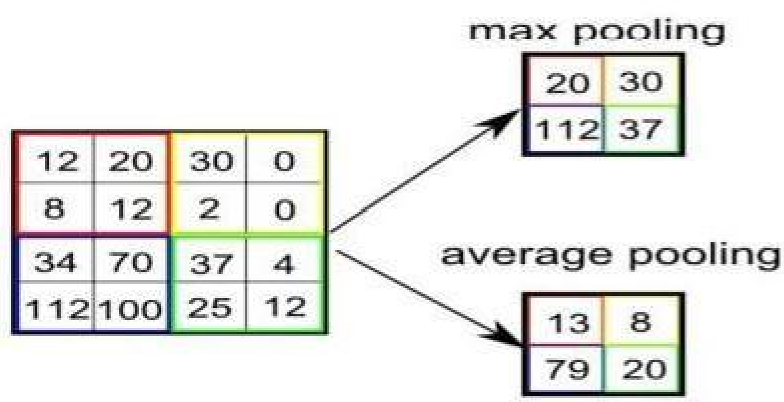
IDE: Google Colab

Theory:

Convolutional Neural Networks (CNN) are complex feed forward neural networks. CNNs are used for image classification and recognition because of its high accuracy. There are three types of layers in a convolutional neural network: i. Convolutional layer ii. Pooling layer iii. Fully connected layer Each of these layers has different parameters that can be optimized and performs a different task on the input data.

What is Pooling Layer?

Pooling layer is responsible for reducing the spatial size of the Convolved Feature. This is to decrease the computational power required to process the data through dimensionality reduction. There are two types of Pooling i. Average Pooling. ii. Max Pooling Max Pooling returns the maximum value from the portion of the image covered by the Kernel. On the other hand, Average Pooling returns the average of all the values from the portion of the image covered by the Kernel. Max Pooling also performs as a Noise Suppressant. It discards the noisy activations altogether and also performs de-noising along with dimensionality reduction. On the other hand, Average Pooling simply performs dimensionality reduction as a noise suppressing mechanism. Hence, we can say that Max Pooling performs a lot better than Average Pooling.



What is Convolutional Layer?

Convolutional layers are the major building blocks used in convolutional neural networks. A convolution is the simple application of a filter to an input that results in an activation. Repeated application of the same filter to an input results in a map of activations called a feature map, indicating the locations and strength of a detected feature in an input, such as an image. A convolutional layer contains a set of filters whose parameters need to be learned. The height and weight of the filters are smaller than those of the input volume. Each filter is convolved with the input volume to compute an activation map made of neurons.

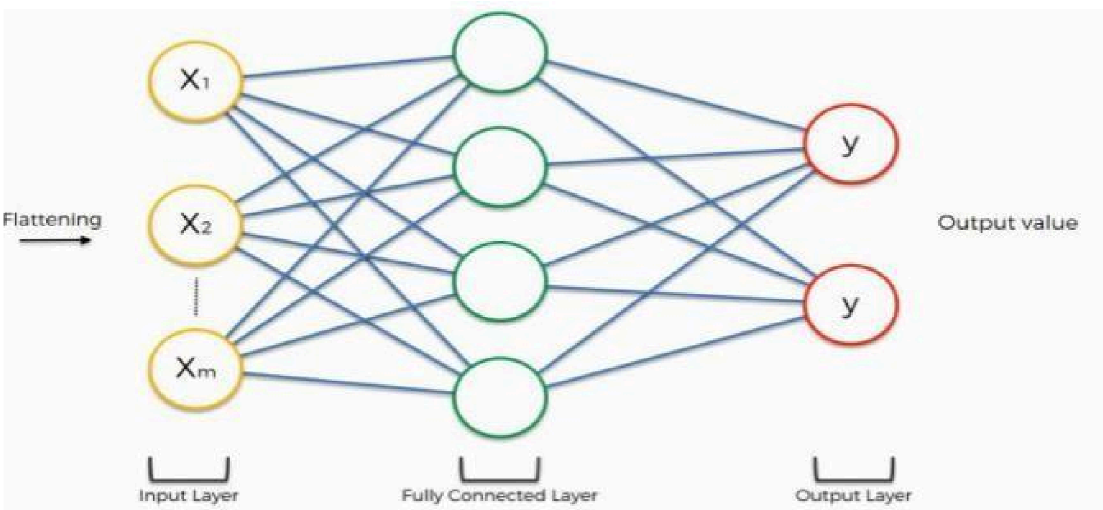
What is Fully Connected Layer?

A fully connected layer that takes the output of convolution/pooling and predicts the best label to describe the image We have three layers in the full connection step i. Input layer ii. Fully-connected layer iii. Output layer

Input Layer: It takes the output of the previous layers, "flattens" them and turns them into a single vector that can be an input for the next stage.

Fully Connected Layer: It takes the inputs from the feature analysis and applies weights to predict the correct label.

Output Layer: It gives the final probabilities for each label.



ReLU Layer: ReLU is an activation function. Rectified Linear Unit (ReLU) transform function only activates a node if the input is above a certain quantity, while the input is below zero, the output is zero, but when the input rises above a certain threshold, it has a linear relationship with the dependent variable. The main aim is to remove all the negative values from the convolution. All the positive values remain the same but all the negative values get changed to zero.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Normalize the data
5. Pre-process the data
6. Visualize the Data

7. Write the CNN model function
8. Write the Cost Function
9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Observe the cost function vs iterations learning curve

Program (Code):

```
import tensorflow as tf
from tensorflow.keras import datasets,
layers, models import matplotlib.pyplot
as plt

(X_train, y_train), (X_test, y_test) =

datasets.mnist.load_data() X_train, X_test =

X_train / 255.0, X_test / 255.0

X_train =
X_train.reshape(-1,28
,28,1) X_test =
X_test.reshape(-1,28,
28,1)

model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu',
    input_shape=(28,28,1)),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64, (3,3),
    activation='relu'),
    layers.MaxPooling2D((2,2)),

    layers.Flatten(),
    layers.Dense(64,
    activation='relu'),
    layers.Dense(10,
    activation='softmax
    ')
```

```
])

model.compile(optimizer='adam',
          loss='sparse_categorical_crossen
          tropy', metrics=['accuracy'])

history = model.fit(X_train, y_train, epochs=5, validation_data=(X_test,

y_test)) model.summary()

plt.plot(history.histor
y['accuracy'])
plt.plot(history.histor
y['val_accuracy'])
plt.title('Accuracy vs
Epochs')
plt.legend(['Train',
'Validation'])
plt.show()

plt.plot(history.histor
y['loss'])
plt.plot(history.histor
y['val_loss'])
plt.title('Loss vs
Epochs')
plt.legend(['Train',
'Validation'])
plt.show()
predictions =


model.predict(X_test)


import numpy as np


plt.imshow(X_test[0].reshape(28,28), cmap='gray')
plt.title("Predicted: " +
str(np.argmax(predictions[0]))) plt.show()
```

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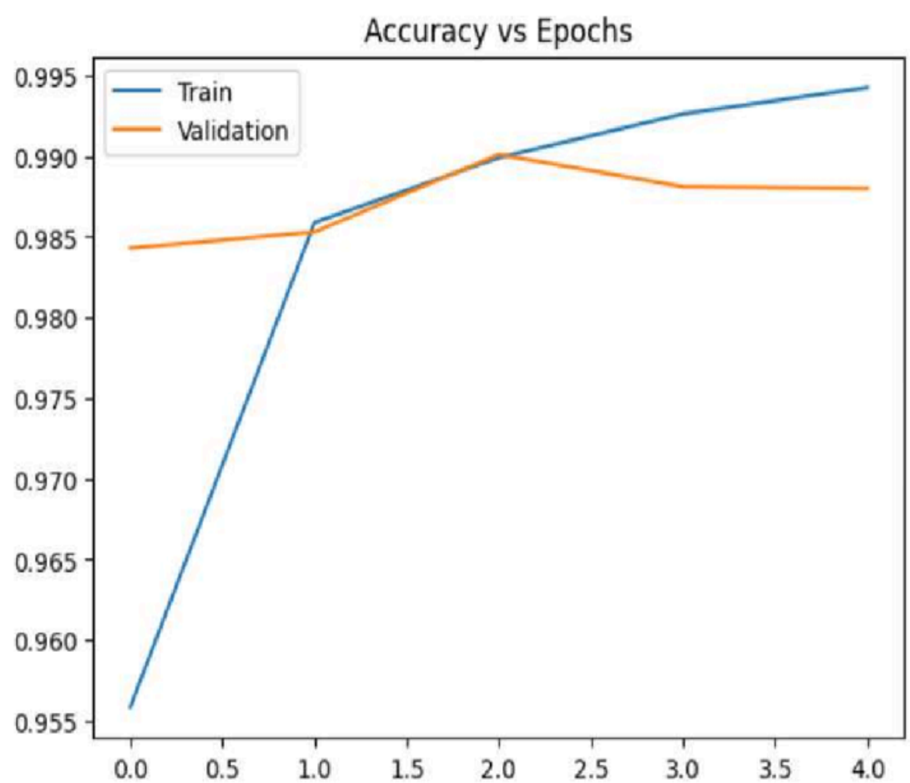
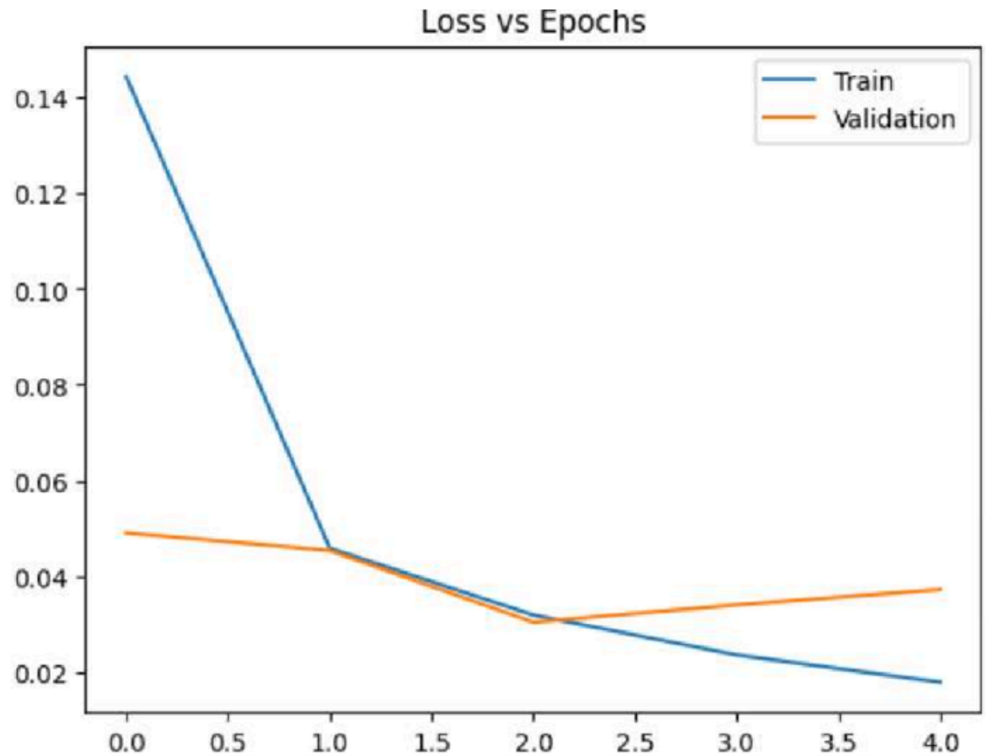
Results:

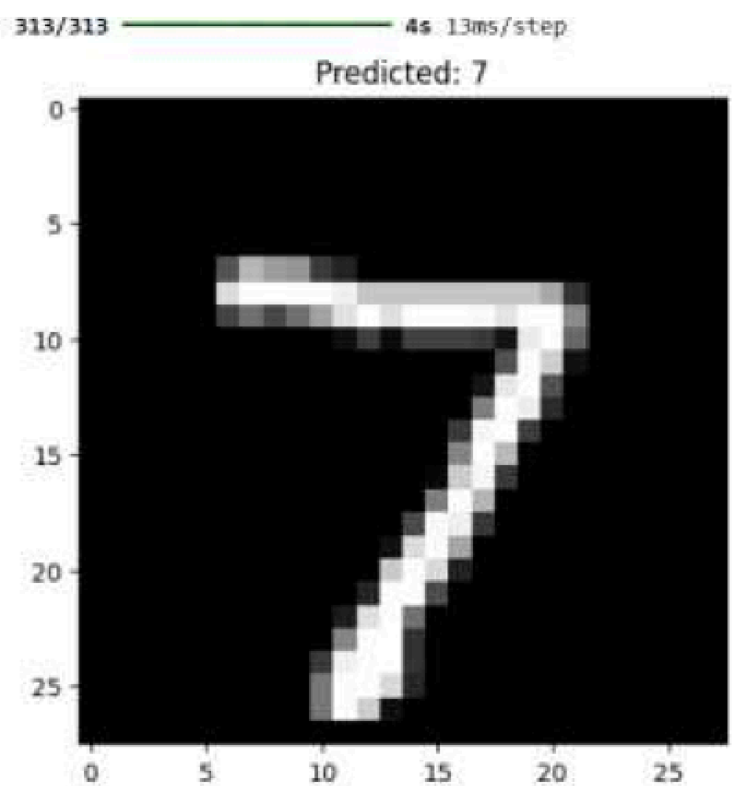
Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 64)	102,464
dense_1 (Dense)	(None, 10)	650

Total params: 365,792 (1.40 MB)
Trainable params: 121,930 (476.29 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 243,862 (952.59 KB)




Loss vs Epochs
Train
Validation
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0


313/313 4s 13ms/step
Predicted: 7
0
5
10
15
20
25

Pre Lab Exercise:

a. **What is the Convolution process? Explain giving examples.**
The Convolution process is a mathematical operation used to extract features from an input image. It involves a small matrix called a Kernel (or Filter) sliding across the input image pixels.
- The Operation: At each position, an element-wise multiplication is performed between the kernel and the overlapping part of the image, and the results are summed up to produce a single value in the Feature Map.
- Example: Imagine an image representing a handwritten "7". A vertical line detector kernel will have high values in its middle column. As it slides over the vertical stroke of the "7", the multiplication results in a high sum, "highlighting" that vertical edge in the output feature map.

b. What are the different layers in the CNN model?
A standard CNN architecture consists of three main types of layers:
- Convolutional Layer: The core building block where kernels extract features like edges, textures, or shapes.
- Pooling Layer: Used to reduce the spatial dimensions (width and height) of the feature maps.
- Fully Connected (FC) Layer: Located at the end of the network. It flattens the 2D feature maps into a 1D vector and uses them to classify the image into categories (e.g., "Tree" vs. "Leaf").

c. What is the requirement of the pooling layer in the CNN model?
The Pooling layer is crucial for making the network efficient and robust. Its main requirements include:
- Dimensionality Reduction: By downsampling (e.g., Max Pooling), it reduces the number of parameters and computations in the network, which helps prevent overfitting.
- Invariance: It makes the model "translation invariant." This means if a feature (like a leaf) shifts slightly in the image, the pooling layer still captures its presence, making the model more flexible.
- Feature Selection: Max Pooling specifically selects the most prominent feature in a local neighborhood, discarding less relevant information.

d. What is the requirement of the use of ReLU activation function after the convolution step?
After a convolution operation, the output is linear. However, real-world data (like images) is highly complex and non-linear. The Rectified Linear Unit (ReLU) is used for:
- Introducing Non-Linearity: The formula for ReLU is $f(x) = \max(0, x)$. By turning all negative pixel values to zero, it allows the network to learn complex patterns that a linear system cannot represent.
- Sparsity: It activates only a subset of neurons (those with positive values), which makes the network computationally efficient.
- Mitigating Vanishing Gradients: Unlike functions like Sigmoid or Tanh, ReLU does not saturate for high positive values, allowing the model to train faster during backpropagation.

Post Lab Exercise:

a. **Why CNN is preferred over ANN for images**
CNN is preferred because:
- It captures spatial features
- Requires fewer parameters

- Better accuracy in image tasks
- Automatically extracts features

b. Can CNN be applied over Text data? If yes, then how. If no, then why?

Yes, CNN can be applied to text by:

a. Converting text into vectors (word embeddings)
b. Applying convolution to detect patterns (like phrases)

c. What is the role of the dropout layer?

Dropout randomly disables neurons during training to:

- Prevent overfitting
- Improve generalization
- Make model robust

d. What will happen if maxpooling is replaced with minpooling?

- MinPooling selects smallest values
- Important features may be lost
- Model performance decreases
- Not commonly used

Conclusion

The experiment successfully demonstrates that **Convolutional Neural Networks (CNNs)** are highly effective for image classification tasks due to their ability to automatically extract spatial features. By utilizing a structured layered architecture, the model achieves high accuracy while managing computational efficiency.

Key Technical Findings:

- **Layer Synergy**: The combination of **Convolutional layers** for feature detection, **ReLU layers** for introducing non-linearity and removing negative values, and **Pooling layers** for dimensionality reduction creates a robust framework for processing visual data.
- **Pooling Efficiency**: Max Pooling was found to be superior to Average Pooling because it not only reduces computational power through dimensionality reduction but also acts as a **noise suppressant** by discarding noisy activations.
- **Dimensionality Management**: The use of a **Fully Connected Layer** is essential at the end of the process to "flatten" the abstract feature maps into a single vector, allowing for the final probability prediction of image labels.
- **Performance vs. ANN**: CNNs are preferred over standard Artificial Neural Networks (ANN) for images because they require fewer parameters and are specifically designed to capture the spatial hierarchy of pixels.